AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

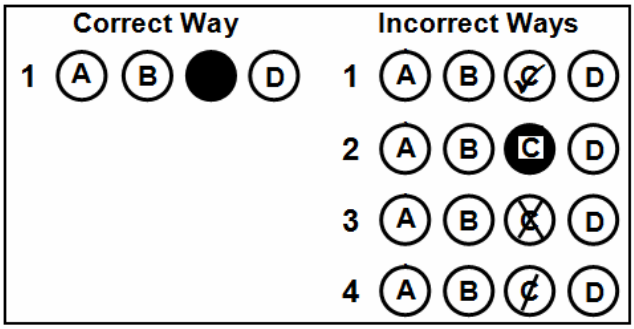
MAY 2013

Chemistry Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. A physiologist took a blood sample from a diabetic patient and examined the composition and amount of each component present in the sample using different techniques and instruments.

   His study is related to the field of

   A. nuclear chemistry.
   B. physical chemistry.
   C. inorganic chemistry.
   D. analytical chemistry.

2. The empirical formula of a compound is CH_2O and its molecular mass is 90. What will be its molecular formula?

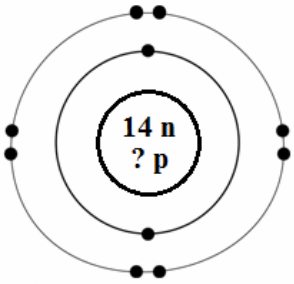
   A. CH_2O
   B. $C_2H_4O_2$
   C. $C_3H_6O_3$
   D. $C_4H_8O_4$

3. Which of the following is an example of a displacement reaction?

   A. $Cl_{2(g)} + 2NaBr_{(aq)} \rightarrow 2NaCl_{(aq)} + Br_{2(g)}$
   B. $H_2SO_{4(aq)} + 2LiOH_{(aq)} \rightarrow Li_2SO_{4(aq)} + 2H_2O_{(l)}$
   C. $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$
   D. $2KClO_{3(s)} \rightarrow 2KCl_{(s)} + 3O_{2(g)}$

4. How many protons are there in an aluminium ion $\left(Al^{3+}\right)$?

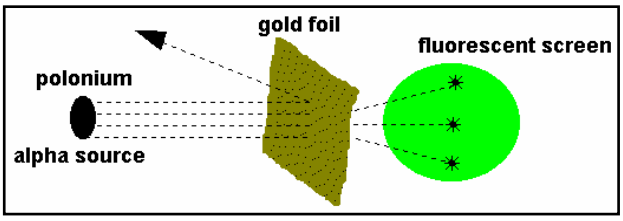
   A. 11
   B. 12
   C. 13
   D. 14



Structure of Aluminium Ion

5. In Rutherford's gold foil experiment, the complete re-bouncing of alpha particles showed that the nucleus of an atom is

   A. hard and dense.
   B. dense and hollow.
   C. hard and negative.
   D. small and neutral.



6. Which of the following electronic configurations represents an element of group IA?

   A. $1s^2, 2s^2, 2p^5$
   B. $1s^2, 2s^2, 2p^6$
   C. $1s^2, 2s^2, 2p^6, 3s^1$
   D. $1s^2, 2s^2, 2p^6, 3s^2, 3p^4$

7. Which of the following pairs of molecules has double covalent bonds in their structures?

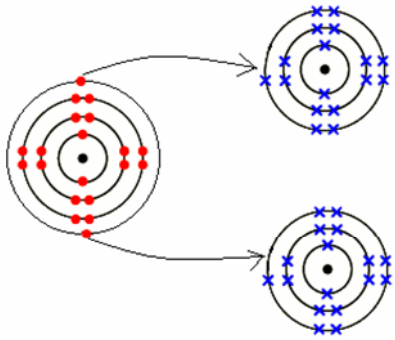
A. O_2 and HCl
B. NH_3 and N_2
C. H_2O and C_2H_2
D. CO_2 and C_2H_4

8. Which of the following compounds shows hydrogen bonding?

A. H_2S
B. CH_3OH
C. C_6H_6
D. CH_3Cl

9. Which of the following ionic compounds is represented by the given dot and cross structure?

A. Na_2O
B. K_2O
C. $CaCl_2$
D. $MgCl_2$



10. An anion (X^{2-}) has an atomic mass 32 and a total of 18 electrons. To which period and group will the element belong?

	Period	Group
A	6	III
B	3	VI
C	2	IV
D	4	II

11. An element belonging to group VI of the periodic table will form an ion

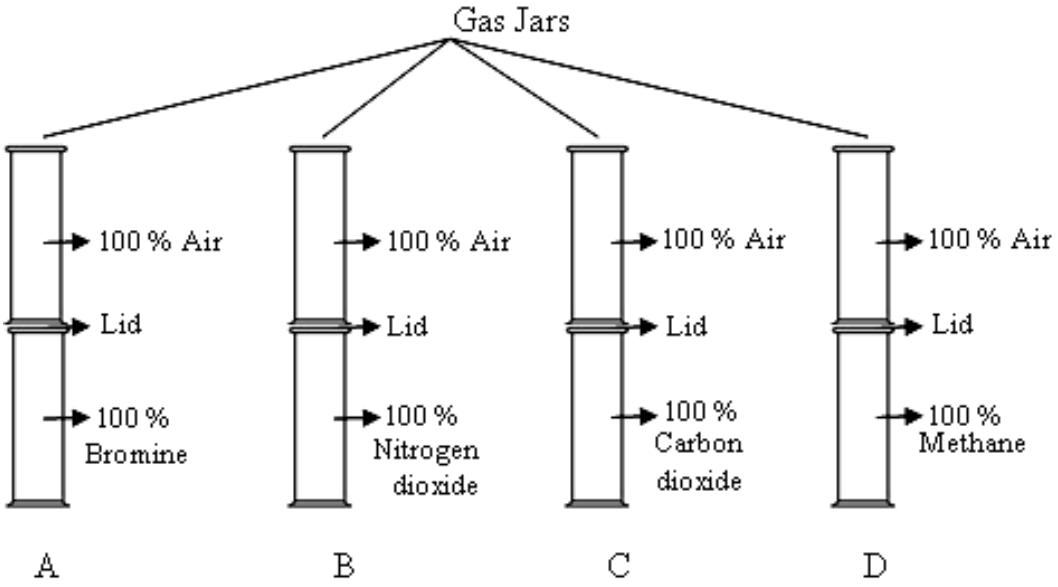
A. X^{+6}
B. X^{-6}
C. X^{2+}
D. X^{2-}

12. Cesium (Z = 55) loses electrons more readily than potassium (Z = 19). This is because of greater

A. shielding effect.
B. ionization energy.
C. electronegativity.
D. electron affinity.

13. Which of the following gases will diffuse most rapidly if the lid is removed?

[Hint: Atomic masses for Br = 80 amu, N = 14 amu, O = 16 amu, C = 12 amu and H = 1 amu]



14. If you put a drop of ethanol on a watch glass, what happens to the drop after a while?

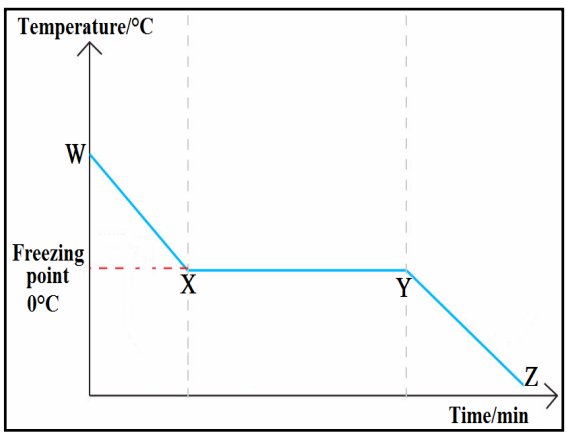
The drop of ethanol will

A. freeze.
B. evaporate.
C. condense.
D. diffuse.

15. The given graph shows the freezing point of water.

What is the state of water between X and Y?

A. Solid
B. Liquid
C. Liquid + Solid
D. Solid + Gas



16. Two persons went to a shop and asked for fizzy drinks. They were offered two bottles, one chilled and the other warm. The chilled fizzy drink tasted better than the other. What could be the reason?

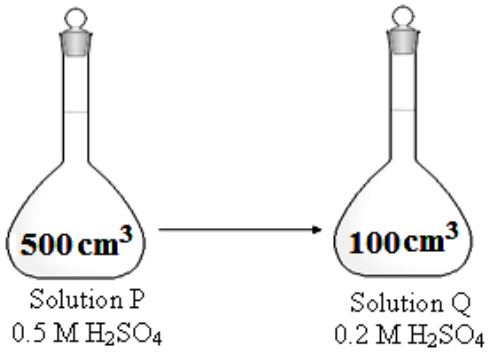
A. Less carbon dioxide is lost at higher temperature.
B. Less carbon dioxide is lost at lower temperature.
C. More carbon dioxide is lost at higher pressure.
D. Less carbon dioxide is lost at lower pressure.

17. Which of the following types of solution will be formed by the smoke of a cigarette in air?

A. A gas into gas
B. A gas into solid
C. A solid into gas
D. A liquid into gas

18. How much volume of solution P is used for making solution Q?

A. 0.04 cm^3
B. 40 cm^3
C. 250 cm^3
D. 200 cm^3



19. Which of the following is an oxidizing agent in the given redox reaction?

$$CuO_{(s)} + H_{2(g)} \rightarrow Cu_{(s)} + H_2O_{(g)}$$

A. CuO
B. H_2
C. Cu
D. H_2O

20. Which of the following represents the correct oxidation states of carbon in the given compounds?

	CO	**CO_2**	**H_2CO_3**
A	0	+2	+1
B	+1	+2	+1
C	+2	+4	+1
D	+2	+4	+4

21. Which of the following statements about rusting of iron is FALSE?

A. Its formula is $Fe_2O_3.nH_2O$.
B. It is a redox chemical reaction.
C. It is accelerated in the presence of acidic medium.
D. It occurs in the presence of distilled deoxygenated water.

22. Which of the following characteristics of calcium makes it more reactive than magnesium?

	Ionization Energy	**Effective Nuclear Charge**
A	Low	High
B	Low	Low
C	High	Low
D	High	High

23. All of the following properties of gold make it useful for making a space craft EXCEPT it is

A. the most reactive metal.
B. able to reflect light.
C. malleable and ductile.
D. resistant to corrosion.

24. Which of the following is an alkaline earth metal?

A. Sodium
B. Radium
C. Rubidium
D. Cesium

25. Sodium metal is kept under paraffin oil because it

A. is soft.
B. has low density.
C. has low melting point.
D. is highly electropositive.

END OF PAPER

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